

Curtin University, Semester 2, 2026

ECOM5005 Tutorial 7B

Confidence Interval Estimation

Preparation expectation. Attempt every question before class. Show your formula, identify the correct critical value, and write a sentence interpreting every interval in the context of the business problem.

Learning Outcomes

1. Construct confidence intervals for a population mean when the population standard deviation is known.
2. Construct and interpret Student t confidence intervals when the population standard deviation is unknown.
3. Construct confidence intervals for a population proportion and check the large-sample conditions.
4. Determine required sample sizes for estimating a mean or a proportion to a specified margin of error.
5. Explain how confidence level, sample size and variability affect confidence-interval width.
6. Use Excel to carry out a complete confidence-interval analysis using a real-world business dataset.

PART 1 - Applied tutorial problems

Question 1 - Solar-farm battery capacity (σ known)

An Australian renewable-energy operator monitors usable capacity of battery modules after three years in service. From long-run calibration studies, the population standard deviation is treated as known at $\sigma = 12$ kWh. A simple random sample of $n = 64$ modules has mean capacity $\bar{x} = 84.6$ kWh.

- (a) Construct a 90% confidence interval for μ .
- (b) Construct a 95% confidence interval.
- (c) Construct a 99% confidence interval.
- (d) Compare the three interval widths and explain why they differ.
- (e) The engineering target is a population mean above 80 kWh. What does the 99% interval suggest about this target? State the inference carefully.

Question 2 - Airport baggage-drop processing time (σ unknown)

An airport operator is evaluating a new automated baggage-drop system. For a random sample of 18 passengers, the time from scanning a bag tag to conveyor acceptance has $\bar{x} = 31.8$ seconds and sample standard deviation $s = 6.4$ seconds. The population standard deviation is unknown.

- (a) Identify the appropriate confidence-interval distribution and degrees of freedom.
- (b) Construct a 95% confidence interval for the true mean processing time.
- (c) The service target is a mean processing time below 35 seconds. What does your interval suggest?
- (d) Because n is small, what assumption matters and what plots would you examine before relying on the interval?

Question 3 - Fintech customer retention (population proportion)

A fintech firm wants to estimate 90-day customer retention after launching a new budgeting app. In a random sample of 250 new subscribers, 146 are still active after 90 days.

- (a) Calculate the sample proportion and verify the large-sample conditions for a normal confidence interval.
- (b) Construct a 95% confidence interval for the population retention proportion π .
- (c) Interpret the interval in percentage terms.
- (d) Marketing claims that 60% of new subscribers are retained after 90 days. Is 60% compatible with your

Question 4 - Cold-chain audit sample-size planning (mean)

A cold-chain logistics company wants to estimate the mean delay, in minutes, caused by temperature-control inspections at an international distribution hub. Historical records suggest $\sigma \approx 18$ minutes.

- How many shipments are required to estimate the mean delay to within ± 3 minutes with 95% confidence?
- How many are required with 99% confidence?
- Explain why the 99% design needs a larger sample.
- If the company can afford to observe only 150 shipments, what margin of error would it achieve at 95% confidence, using $\sigma = 18$?

interval? Explain without conducting a separate hypothesis test.

Question 5 - AI-helpdesk escalation survey (sample size for a proportion)

A software company is planning a customer survey to estimate the proportion of AI-helpdesk conversations that are escalated to a human agent. A pilot study suggests $\pi \approx 0.18$. Management wants a 95% confidence interval with margin of error no more than ± 2.5 percentage points.

- Using the pilot estimate 0.18, determine the required sample size.
- Recalculate using the conservative choice $\pi = 0.50$.
- Explain why $\pi = 0.50$ gives the largest required sample.
- If only 1,000 conversations can be reviewed, calculate the approximate 95% margin of error using $p = 0.18$ and, separately, using the conservative $p = 0.50$.

PART 2 - Data exercise: Restaurant tips and confidence intervals

Dataset. Use the accompanying Excel file “Restaurant_Tips_Data.xlsx”. The data contain 244 restaurant transactions recorded by one food server over about two and a half months. The workbook includes a fixed random sample of 60 observations for the tutorial.

Source: Bryant, P. G. & Smith, M. A. (1995), Practical Data Analysis: Case Studies in Business Statistics. Dataset description: <https://iowabiostat.github.io/data-sets/tips/tips.html>

Use Sample_60 for your calculations and Analysis_Workspace for your answers. Do not use Full_Data until the final benchmarking question.

- Compute descriptive statistics for total_bill and tip: sample size, mean, median, sample standard deviation, minimum, Q1, Q3, maximum, skewness and kurtosis.
- Construct and interpret a 95% Student t confidence interval for the mean total bill. Show the standard error, degrees of freedom, critical t value and margin of error.
- Construct and interpret a 99% Student t confidence interval for the mean tip.
- Estimate the proportion of sampled dining parties that include smokers. Check the large-sample conditions and construct a 95% confidence interval for the population proportion.
- Suppose a future study wants the mean total bill estimated to within $\pm \$2$ at 95% confidence. Use the sample standard deviation as an estimate of σ and determine the required sample size.
- Suppose a future study wants the smoker proportion estimated to within ± 5 percentage points at 95% confidence. Find the required sample size using the sample proportion, then repeat using the conservative value $p = 0.50$.
- Create a histogram of total_bill for Sample_60 and comment on shape, centre and any large bills. Explain why the t interval is still reasonable with $n = 60$ even if the sample is not perfectly normal.
- Only after completing (a)-(g), use Full_Data as an observed benchmark. Compare the full-data mean total bill, mean tip and smoker proportion with your sample confidence intervals. Explain what this check teaches about interval estimation.

(i) State the population to which these intervals can reasonably be generalized. Explain why it would be inappropriate to claim that the results describe all restaurants or all customers.

Class Discussion

Bring your completed Excel workbook to class. Be prepared to explain which confidence-interval formula you used, why it was appropriate, how you selected the critical value, and what the interval does - and does not - imply about the population.

PART 3 - Pearson MyLab

1. Complete **Quiz #4** on MyLab which has been released on **Tuesday (22-Sep-2026)** and will be **due by the Tuesday on 6-Oct-2026 at 5:00pm AWST**.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email to the teaching team explaining the issue. It is important that you complete MyLab tasks on a weekly basis since the final e-test in this unit will be completed using MyLab.